Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

the de Broglie wavelength of a particle can be calculated using the formula Î» = h / sqrt (2 m E), where Î» is the de Broglie wavelength , h is the Planck constant , m is the mass of the particle , and E is its kinetic energy .

First , convert the kinetic energy from eV to Joules . 1 eV = 1 . 602 x 10 ^ - 19 J . So the kinetic energy is 1 . 602 x 10 ^ - 19 J .

The mass of an electron is approximately 9 . 11 x 10 ^ - 31 kg .

The Planck constant is approximately 6 . 626 x 10 ^ - 34 J s .

Now we can plug in the values into the formula to get Î» = (6 . 626 x 10 ^ - 34) / sqrt (2 * 9 . 11 x 10 ^ - 31 * 1 . 602 x 10 ^ - 19) = (6 . 626 x 10 ^ - 34) / sqrt (2 * 9 . 11 x 10 ^ - 31 * 1 . 602 x 10 ^ - 19)

Simplify the expression to find the de Broglie wavelength .

Then convert the wavelength to cm and check the answer choice options .

</think><answer> B . 5 . 92 ÃĹ 10 ^ 7 cm </answer>